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@page { margin: 14mm 12mm; } body { font-family: "DejaVu Sans",
Arial, sans-serif; font-size: 10.5pt; line-height: 1.34; color: #1a2733;
max-width: 100%; margin: 0; } h1 { font-size: 19pt; color: #12324b;
border-bottom: 2px solid #12324b; padding-bottom: 3px; } h2 { font-
size: 14pt; color: #12324b; border-left: 4px solid #12324b; padding-
left: 7px; margin-top: 16px; } h3 { font-size: 12pt; color: #234;
margin-top: 12px; } table { border-collapse: collapse; width: 100%;
margin: 8px 0; font-size: 9pt; } th, td { border: 1px solid #b7c2cc;
padding: 3px 6px; text-align: left; vertical-align: top; } th {
background: #12324b; color: #fff; } tr:nth-child(even) td {
background: #f2f5f7; } code { background: #eef1f4; padding: 1px
3px; border-radius: 3px; font-size: 8.8pt; } pre { background:
#0f1b26; color: #dfe7ee; padding: 8px 10px; border-radius: 5px; font-
size: 8.4pt; overflow-x: auto; white-space: pre-wrap; } pre code {
background: transparent; color: inherit; }
```

ORAN_VIP_ARCHITECTURE

O-RAN VIP Suite — Architecture

Lane-1 UVM-role mapping (plain-SV subset)

seq_item = _hdr_t (64-bit protocol header, packed struct).
sequence/CRV = Z3 seeded generator (gen_stim.py) — pos/-neg/-
directed. driver = procedural field-apply in TB. monitor = COVROW
emit. scoreboard = 3 checks: runtime legality self-check, pack-vs-
golden cross-check, round-trip identity. coverage = Python COV-###
engine (functional) + verilated coverage.dat (code). reporter = UVM-
format summary; adjudicate.py = truth authority.

Cross-constraint solving

Z3 enforces cross-field legal bounds the SV simulator cannot
(e.g. start_prb+num_prb<=275, start_symbol+num_symbol<=14,
num_layers<=num_ports<=ant_cfg, comp_width<=iq_width,
rank<=num_layers). Proven: 0 violations over 900+ txns per
constrained component.

Lane-2 delta (future, licensed sim / DSim — GAP-ORAN-002)

Native randomize()+{} replaces Z3 pre-gen; native covergroup
(cross+illegal_bins) replaces Python engine; concurrent SVA in bind
replaces procedural checkers; UVM factory/config_db/
sequences/uvm_scoreboard replace plain-SV roles. Same FR/COV/SVA
IDs; same result.json contract.

Coverage model

Functional: per-field COV-### bins (enum/bool/wrap/bucket4). Code:
line 100%, branch (DUT) 100%, 6 TB-defensive branches/comp
waived. Toggle reported (not a GATE-8 criterion).